

LITERATURE REVIEW – DOMAIN & MACHINE LEARNING

1. CORE DOMAIN CONCEPTS

AI and Machine Learning foundation

- Supervised regression: learning of functions from labelled inputs (V, I, T, power, etc.) to continuous numerical targets (SoC, remaining time) based on historical data
- Ensemble tree methods (Random Forest, XGBoost): combination of decision trees trained on subsets of the data, where individual RF trees are averaged (bagging), while the boosting-type algorithm XGBoost sequentially trains trees to correct predecessor errors
- Artificial Neural Network (ANN) / deep sequence models (LSTM): non-linear and temporal pattern learning from raw or engineered features, typically requiring more data than tree ensembles but being able to capture sequence effects that the latter cannot
- MAE, RMSE, R^2 : the three metrics used throughout this comparison, Mean Absolute Error and Root Mean Squared Error in the target's own units (percent SoC, minutes), and R^2 as the proportion of variance explained, each calculated separately for the SoC target and the remaining-time target.
- Session-level, leakage-safe splitting: dividing the data at the charging session level rather than the row level, to avoid data leakage, the very same method the original studies used to obtain a realistic, non-inflated performance estimate.

Battery Analysis

- SoC (state-of-charge): percentage of the battery that can still be charged, the main target to be regressed, this is not directly measurable, and has

historically been estimated using Coulomb counting or circuit models by tracking the integrated current.

- Remaining-charging-time: the other regression target, representing how long it will take to fully (or to a target) charge the battery, which can be framed as a “time-to-end” within a single session.
- Coulomb counting: integration of current measurements to estimate charge, which is simple to implement but accumulates errors over long sessions.
- Equivalent-circuit / Kalman-filter models: physics-based approaches that model the internal workings of the battery (using RC circuits or state-space filters), which require tuning as the internal resistance and capacity change with aging.

2. Public Dataset & Prior Systems

a. Dataset:- NASA PCoE

Content:- 4 commercial 18650 cells, full charge–discharge cycles under controlled lab conditions, capacity + impedance tracked to end-of-life.

Used for:- SOH / degradation trend modelling; small, clean benchmark for quick prototyping.

Suitable for:- Good for pipeline smoke-testing, but too small and too clean to reflect real charger sensor noise.

b. Dataset:- CALCE (CS2 / CX2)

Content:- Prismatic LCO cells, standard CC/CV charge protocol, varying discharge/aging conditions, University of Maryland.

Used for:- Cross-validating degradation and voltage-curve models against a second lab source.

Suitable for:- Useful as an external sanity check once a model is trained on the project's own session data.

c. Dataset:- MIT / Stanford Fast-Charging Datasets

Content:- 124–135 commercial LFP/graphite 18650 cells, 224 distinct fast-charging protocols, cycled to end-of-life at 30°C.

Used for:- Large-scale ML on fast-charging behaviour and cycle-life prediction — the closest public analogue to charging-time prediction.

Suitable for:- Best public reference for structuring a charging-time target and feature set if the project's own data is limited.

d. Dataset:- HNEI (Hawaii Natural Energy Institute)

Content:- 51 NMC-LCO 18650 cells, cell-to-cell degradation variability, reference performance tests.

Used for:- Studying variability across nominally identical cells — relevant to generalisation across a fleet/session set.

Suitable for:- Illustrates why session-level (not row-level) splitting matters: cells/sessions differ systematically.

e. Dataset:- Real-world DC fast-charging telemetry (909k sessions, Nature Comms 2025)

Content:- Multi-vehicle, multi-station charging profiles and durations collected 2011–2024.

Used for:- Directly analogous to the remaining-charging-time target; shows what accuracy is achievable at very large scale.

Suitable for:- Sets an upper-bound expectation (90–95% duration accuracy) — useful for judging whether a smaller project dataset is data-limited.

3. Literature Review

Study	Approach	Key Takeaway
Li et al. — RF for SoC (IEEE)	Random Forest regression vs. BP neural network	RF beat NN baseline on dynamic profiles — confirms tree ensembles suit this problem

RF vs. SVM vs. GB (IEEE 2024)	Compared across 3 temperatures	RF is robust and temperature-stable — justifies RF as core baseline
DSA-Optimised RF (MDPI 2024)	Metaheuristic-tuned RF	Plain RF is weakest at start/end of charge cycle — informs error analysis by SoC band
RF+XGBoost Voting Ensemble (2026)	Blended RF + XGBoost	Ensemble beat any single model — fallback if individual models plateau
XGBoost for SOH (Nature Sci. Reports, 2025)	Gradient boosting on V/I/temp/resistance	XGBoost learns non-linear degradation well — transfers to SoC/time
XGBoost–RF Fusion, Lei (2024)	Fused XGB + RF outputs	Fusion beat linear & single models on non-linear voltage curve
Two-Stage Cascaded LSTM (2025)	LSTM forecasts voltage → LSTM estimates SoC	Sequence models help with temperature-distorted curves — possible extension
DL for DC Fast-Charging Duration (2025)	Deep learning, real-world telemetry	Closest match to time prediction target — field less explored than SoC

- **Random Forest** is an effective baseline, but its accuracy consistently drops at the extreme start and end of charge/discharge cycles—requiring error analysis across specific SoC bands.
- **XGBoost** effectively handles non-linear voltage and degradation curves on tabular data, justifying its role as a core model alongside Random Forest.
- **Fusing Random Forest and XGBoost** delivers substantial performance gains (up to 47% error reduction) and serves as a strong backup strategy if single models plateau.
- **Recurrent/Sequence models (LSTMs)** are only worth their added complexity if tree models fail to account for temperature-induced voltage distortions.
- **Remaining charging time prediction** is significantly under-researched compared to SoC estimation, leaving a distinct gap in the existing literature

4. Candidate ML Methods

- **Linear / Simple Baseline:-** Establishes a reference score; makes later gains from complexity measurable and honest.
- **Random Forest Regressor:-** Handles non-linear voltage/current interactions well; robust to noise and outliers with little tuning.
- **XGBoost:-** Gradient boosting typically wins on tabular sensor data; strong accuracy-to-effort ratio, used in most charging-time literature.

- **ANN (Optional):**- Tried only if the tree-based models underfit sequential/non-linear charging dynamics; adds capacity, needs more data & tuning.